CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (ME-AMT) Examination

Dec. 2013- Jan. 2014

ME731 Casting and Welding Processes

Date: 01.01.2014, Wednesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer following multiple choice type questions. There could be more than one correct answer. Write all correct choices in answer sheet. **[10]**

(i) In a cube casting made with split pattern with horizontal parting line, which of the following option leads to undercut:
(p) Through vertical hole (q) Through horizontal hole at parting line
(r) Blind vertical hole in drag (s) Blind horizontal hole in drag

(ii) Liquid metal fluidity depends on:
(p) Pouring temperature (q) Mold surface roughness
(r) Alloy composition (s) All of the above

(iii) During solidification, phase change occurs nearly at constant temperature in:
(p) Pure metals (q) Hyper-eutectic alloys
(r) Hypo-eutectic alloys (s) Eutectic alloys

(iv) In general for die casting, high resistance to heat transfer is offered:
(p) In solidified metal (q) In mold material
(r) At metal-mold interface (s) At mold-air interface

(v) In horizontal multi-gate system smallest discharge occurs at:
(p) Farthest gate to the sprue (q) Nearest gate to the sprue
(r) Equal discharge occurs at each gat (s) None of the above

(vi) Metal contraction during casting solidification (phase-change) is compensated by:
(p) Providing additional feeder (q) Providing additional gates
(r) Providing additional sprue (s) Shrinkage allowances in pattern

(vii) Arrange the junction in increasing level of difficulties in handling during solidification
1. + junction, 2. L junction, 3. ⋆ junction, 4.T junction, and 5. X junction:
(p) 1-2-3-4-5 (q) 2-4-1-3-5 (r) 2-4-1-5-3 (s) 2-1-5-4-3

(viii) Favorable conditions for mold erosion are:
(p) High impact velocity (q) High density molten metal
(r) High sprue length (s) All of the above

(ix) Which of the following process(es) use a dispensable mold:
(p) Sand casting (q) Centrifugal casting
(r) Shell molding (s) Investment Casting

(x) Exothermic feeder have feeding capacity in the range of:
(p) 10 to 15 percentage (q) 15 to 25 percentage
(r) 25 to 40 percentage (s) 50 to 60 percentage

Q - 2 Hollow steel cylinders with OD=30 cm, ID=10 cm, and length=40 cm are to be produced by sand casting in a mold box with length120 cm, width100 cm, and height 40+40 cm.

(a) For the casting (vertical orientation), design a cylindrical side feeder. **[10]**

(b) Find out how many such feeders are needed. Compute the total feeding efficiency and check if it is too high or low. Is it advisable to connect more than one cavity to a feeder? **[05]**

(c) Calculate the improvement in feeding efficiency and yield, if insulating sleeve of MEF=1.4 is applied. [05]

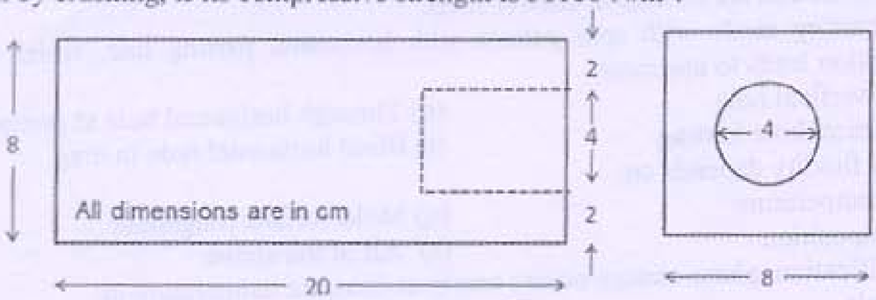
OR

Q - 2 (a) What will be the filling time for a 10 cm cube casting with top, middle, and bottom gating, if sprue height is 20 cm and gate cross-section area is 1 sq. cm. Neglect losses. [10]

(b) Derive the general analytical solution to 1D heat transfer in the castings: $\frac{\partial T}{\partial \tau} = \alpha \frac{\partial^2 T}{\partial x^2}$ [05]

(c) Derive the expression for solidification time of the casting when major resistance to heat transfer is offered at metal-mold interface and inside solidified metal. [05]

Q - 3 For the steel block casting shown below, find the compressive stress on the sand mold caused by the horizontal blind core, assuming core print diameter as 6 cm. Take density of core material 1600 kg/m^3, and density of metal (steel) 8000 kg/m^3. Check if the mold will fail by crushing, if its compressive strength is 50000 N/m^2. [05]



OR

Q - 3 For the steel block casting given above (Q-3), compute shape complexity. [05]

SECTION – II

Q - 4 (a) What is the difference between a consumable electrode and a non-consumable electrode in arc welding? [02]

(b) Explain why is the tungsten inert –gas welding preferred for welding aluminum plates. [03]

(c) How do porosity and slag inclusions occur in welding? [03]

Q - 5 (a) What are the parameters to be controlled in the resistance welding process? [05]

(b) In an arc welding operation, the power source is at 20volt and current is 300A. If the electrode travel speed is 6mm/sec, calculate the cross sectional area of the joint. Take heat transfer efficiency 0.80 and melting efficiency 0.30. Heat required to melt is 10J/mm^3. [07]

OR

Q - 5 (a) Explain process of spray metal and short circuit transfer in GMAW process with sketch. [05]

(b) In a resistance welding of a lap joint or two mild steel sheets of 1.5cm thickness a current of 10,000A is passed for a period of 0.1sec. The effective resistance of the joint is 120micro-ohms. The density of steel is 0.00786g/mm^3 and heat required to melt is 1381J/g. The joint can be considered as a cylinder of 5mm diameter and 2.25mm height. Calculate the % of heat distributed to the surroundings. [07]

Q - 6 Attempt *any three*: [15]

(a) Briefly explain the working principle of the Submerged Arc Welding (SAW) process and mention their applications.

(b) Give a brief comparison between Friction Stir Welding (FSW) and Diffusion Welding (DW) based on at least 10 factors.

(c) Discuss pros and cons of spherical casting feeder.

(d) Discuss various elements of cost involved during cast product manufacturing.

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